

# **UNIT 6**

## **Current Computer Forensic Tools**

# Evaluating Computer Forensic Tool Needs

## Types of Computer Forensic Tools

### A. Hardware Forensic Tools

- Consist of physical devices used for evidence acquisition and protection.
- Range from simple single-purpose components to advanced forensic workstations.
- Examples:
- ACARD AEC-7720WP SCSI-to-IDE Bridge (write blocker)
- Digital Intelligence F.R.E.D. Systems
- DIBS Advanced Forensic Workstations
- Forensic Examination Stations & Portable Units

### B. Software Forensic Tools

- Tools used for disk imaging, analysis, and reporting.
- Categorized into command-line and GUI-based tools.
- Examples:

- Command-line: SafeBack by NTI (disk acquisition)
- GUI-based: ProDiscover, X-Ways Forensics, EnCase, AccessData FTK

## **2. Role of Software Forensic Tools**

- Create exact image files of storage devices.
- Allow reading and analyzing image files as if they were the original disk.
- Used for deep forensic investigation.
- Common analysis tools:
  - ProDiscover
  - EnCase
  - FTK
  - X-Ways Forensics
  - ILook

# Responsibilities / Tasks of Computer Forensic Tools

## 1) Acquisition

### Meaning

- First step in a computer forensic investigation.
- Creates a forensically sound replica of the original drive.
- Protects the original evidence from modification or degradation.

### Subfunctions of Acquisition

- Physical & Logical Copying of Data
- Data Acquisition Formats (Raw, Proprietary, AFF, etc.)
- Verification (check integrity using hash values)
- Command-line Acquisition (e.g., SafeBack)
- GUI-Based Acquisition (e.g., FTK Imager, EnCase)
- Remote Acquisition (capturing data from remote systems)

## Acquisition Tools

- Software tools: FTK, EnCase, X-Ways, ProDiscover
- Hardware imaging tools:
  - Logicube Talon
  - VOOM HardCopy 3
  - Image MASter Solo II
- These hardware tools come with built-in imaging software.

## Note

- Hardware tools can create a duplicate disk without external software.
- However, forensic software is still required for analysis after acquisition.



## **2.) Validation & Discrimination**

### **Meaning**

- Ensures that the copied data is accurate and unchanged (validation).
- Helps separate relevant data from irrelevant data (discrimination).

### **Importance**

- Essential to maintain evidence integrity.
- Essential for efficient analysis—removes unnecessary files.

### **Subfunctions**

- Filtering (remove known-good files)
- Hashing (MD5, SHA-1, SHA-256)
- File Header Analysis (detects file type mismatches)

# Hashing

- Produces a unique hexadecimal value for data.
- Proves that evidence hasn't been altered.
- Used to compare files with known-good hash sets (e.g., NSRL).
- Helps investigators ignore safe files and focus on suspicious files.

## Why Filtering Helps

- Removes OS files, common applications, and trusted documents.
- Reduces workload by eliminating irrelevant data.

## Sector-Level Hashing

- Finds partially deleted fragments in unallocated or slack space.

## Header Analysis

- Detects mismatched file extensions.
- Helps reveal hidden files or disguised malware.

## **3) Extraction**

### **Meaning**

- Most challenging step in forensics.
- Involves recovering, searching, and analyzing all collected data.

### **Subfunctions**

- Data Viewing (files, folders, partitions, metadata)
- Keyword Searching
- Decompression (ZIP, RAR, GZIP, etc.)
- Carving (recover deleted files from raw sectors)
- Decryption (unlock encrypted data)
- Bookmarking / Tagging (mark important evidence)

### **Tools for Extraction**

- |                    |       |
|--------------------|-------|
| • ProDiscover      | FTK   |
| • EnCase           | SMART |
| • X-Ways Forensics |       |



## **Data Viewing Features**

- Shows logical file structures (folders, documents, images, logs).
- Also displays:
  - Allocated data
  - Unallocated space
  - Slack space
- Allows investigators to view raw file content in its natural form.

## **Importance**

- Helps recover deleted items, hidden data, and compressed files.
- Essential for building timelines and linking actions to users.

# Computer Forensics Software Tools

## Overview

- Wide range of forensic software tools available today.
- Some tools are general-purpose for multiple tasks.
- Others are specialized, focusing on specific evidence types like email, memory, network traffic, etc.
- Choice between open-source and commercial tools depends on:
  - Cost
  - Features
  - Capabilities
  - Support

## Functions Performed by Forensic Tools

- Collection of evidence
- Preservation without altering original data
- Analysis of digital artefacts
- Presentation of findings in readable form

# Popular Computer Forensics Tools

## SANS SIFT

- Ubuntu-based Live CD for digital forensics and incident response.
- Supports E01, AFF, RAW (dd) image formats.
- Contains powerful tools like:
  - log2timeline – timeline analysis
  - Scalpel – file carving
  - Rifiuti – recycle bin analysis
- Used for in-depth forensic analysis without installation.

## CrowdStrike CrowdResponse

- Lightweight console-based incident response tool.
- Collects system information:
  - Process lists
  - Scheduled tasks
  - ShimCache data
- Uses YARA signatures to detect malware and indicators of compromise.

## **Volatility**

- Memory forensics framework for RAM dump analysis.
- Extracts:
  - Running processes
  - Network sockets and connections
  - Loaded DLLs
  - Registry hives
  - Process IDs, handles, drivers
- Essential for malware investigation and incident response.

## **The Sleuth Kit (TSK) + Autopsy**

- Open-source toolkit for file system analysis.
- Autopsy provides a GUI on top of TSK.
- Features:
  - Timeline Analysis
  - Hash Filtering
  - File System Browsing
  - Keyword Searches
  - Extendable using modules

## **FTK Imager**

- Tool for imaging and previewing data.
- Examines files/folders from disks, network shares, CDs/DVDs.
- Can:
  - Create SHA-1 / MD5 hashes
  - Recover deleted files
  - Export files from forensic images
  - Mount images for browsing
- Part of AccessData FTK suite.

## **ExifTool**

- Command-line tool for reading and editing metadata.
- Supports a wide variety of file types (especially images).
- Useful for analysing metadata of suspicious files.



## **Free Hex Editor Neo**

- Hex editor capable of loading very large files.
- Useful for:
  - Manual data carving
  - Low-level file modifications
  - Hidden data searching
  - Raw image inspection

## **Bulk Extractor**

- Scans disk images and extracts:
  - Credit card numbers
  - Emails
  - URLs
  - Domains
  - ZIP archives
- Outputs results into text files for manual or automated review.

# Computer Forensics Hardware Tools

## Introduction

- Hardware tools are essential in digital forensics to collect, preserve, and analyze digital evidence.
- Common hardware includes:
  - Cables & adapters
  - Write blockers
  - Cloning devices
  - Cell phone acquisition kits
  - Portable storage devices
  - Forensic workstations

## 1. Computers (Forensic Workstations)

### Role

- Main foundation of every digital forensics lab.
- Forensic tasks require high computational power, large memory, and fast storage.



## **Specifications of a Good Forensic Workstation**

- Multi-core CPUs (Intel i7/i9, Xeon, AMD equivalents)
- Maximum possible RAM
- Fast & large hard disks (SSD + multi-terabyte HDD)
- Meet or exceed recommended hardware requirements of forensic tools

## **FTK (Forensic Tool Kit) Hardware Requirements**

FTK consists of 4 components:

1. Oracle Database
2. FTK UI (Client Interface)
3. Client-side Processing Engine
4. Distributed Processing Engine

## **Storage Requirements**

- Forensics labs handle large volumes of data.
- Modern systems often have 500–700 GB drives.
- Multi-terabyte storage is commonly required due to increasing caseloads.



## **2. Cell Phones & GPS Units**

### **Why Important?**

- Digital forensics is no longer limited to PCs.
- Smartphones and GPS devices contain crucial evidence.

### **Tools**

- Cellebrite UFED
  - Supports 3,000+ devices
- Paraben Device Seizure
  - Supports 4,000+ phones, PDAs & GPS units

### **Hardware Needs**

- Large collection of cables & connectors (mobile devices lack standard ports).
- Many device cables are provided by forensic tool manufacturers.

## **3. Cloning Devices**

### **Purpose**

- Create bit-stream forensic clones of storage media.
- Ensure evidence integrity.

### **Features of Hardware Cloning Tools**

- High-speed multi-disk copying
- Built-in write protection
- Hash authentication
- Disk wiping
- Audit trail generation

### **Examples**

- Tableau Forensic Duplicators
- Image MASter units
- Logicube devices

## 4. Write Blockers

### Purpose

- Prevent data modification on suspect devices.
- Allow read-only access to storage media.

### Types

- Hardware write blockers
- Bridge adapters (IDE, SATA, USB, SCSI)

## 5. Crime Scene Kits

Used Outside the Lab

To collect evidence at the crime scene.

Typical Items in Forensic Field Kits

- Cables & adapters
- Portable write blockers
- Forensically clean USB drives

- Evidence bags & evidence tape
- Permanent markers & labels
- Digital cameras
- Notebook & report forms
- Gloves, tools, and sealable boxes

These kits ensure proper chain of custody and evidence integrity.

# Validating and Testing Forensics Software

After selecting the tools you want to use in a digital forensics investigation, you must ensure that the evidence collected using those tools is accurate, reliable, and acceptable in court.

Therefore, testing and validating forensic software is essential. The National Institute of Standards and Technology (NIST) provides standards, tools, and methods to help test and verify computer forensics tools.

## **NIST and the CFTT Program**

**NIST created the Computer Forensics Tool Testing (CFTT) program to:**

- Evaluate and validate digital forensic tools
- Increase the legal admissibility of evidence
- Ensure tools produce consistent and trustworthy results

## **Steps in Validating and Testing Forensic Tools (According to NIST)**



## **Create Classifications for Forensic Tools**

Forensic tools must be grouped into categories such as:

- Email recovery and tracing tools
- Disk imaging tools
- File carving tools
- Registry analysis tools

This helps in deciding how each category should be tested.

## **Identify Requirements for Each Category**

For every category, the technical requirements must be defined.

Example:

An email forensics tool should be able to extract:

- Sent mails
- Received mails
- Deleted mails
- Attachments

## Create Test Claims (Test Assertions)

These are statements describing what the tool should be able to do. They help verify whether the tool meets the defined requirements.

Example:

“The tool must recover deleted emails from a PST file.”

## Determine Test Cases

Next, you select or design realistic test cases.

These may include:

- Sample drives
- Disk images
- Closed case datasets
- Known evidence sets

The tool is tested on these datasets to see if:

- It extracts the correct information
- It produces the same results as other validated tools

# Develop a Test Strategy

## A test strategy explains:

- How the tool will be tested
- Which features will be checked
- What data will be used
- What the expected results should be

## Report Test Findings (According to ISO 17025 & ISO 5725)

The final test report must be:

- Accurate
- Clear
- Transparent
- Impartial

Repeatability and Reproducibility

- Repeatable results → Same lab, same equipment → same results
- Reproducible results → Different labs, different systems → same results

This ensures the tool is scientifically reliable.



# Email Forensics

Email forensics is the process of examining the origin, path, and content of an email to determine the real sender, receiver, time of transmission, sender's IP, and other hidden details that may serve as digital evidence.

It involves analyzing metadata, attachments, mail headers, logs, and server records to investigate fraud, phishing, spoofing, data leakage, or cybercrimes.

## Techniques Used in Email Forensic Investigation

### Header Analysis

- The email header contains crucial metadata such as sender IP, routing path, timestamps, and mail server information.
- This is the primary and most powerful technique in email forensics.
- By examining header fields, investigators can detect:
  - Email spoofing
  - Phishing attempts
  - Spam campaigns
  - Scams
  - Internal data leakage

- The header reveals the actual source even if the visible sender address is fake.

## **Server Investigation**

- Involves checking mail server logs and retrieving deleted or archived emails from the service provider.
- The investigation may include:
  - Extracting complete mailboxes
  - Analyzing server-side timestamps
  - Verifying IP logs and authentication records
- This method helps when user-level evidence is missing or tampered.

## **Network Device Investigation**

- Sometimes investigators must analyze router, firewall, switch, or proxy logs to trace the real source of an email.
- This technique is used when:
  - Primary server logs are unavailable
  - The ISP or proxy server does not store logs

- Emails are routed through multiple networks
- It is often complex because network devices may not retain complete or consistent logging data.

## **Sender/Mailer Fingerprinting**

- The "Received" headers show the chain of mail servers that handled the message, listed in reverse order.
- Additional fields like X-Mailer or User-Agent reveal:
  - The email software used
  - The version of the application
  - Possible OS or device details
- This helps identify the sender's environment and verify authenticity.

# Roles of Client and Server in E-mail

## Structure of an Email

- An email has two parts:
  - Header → Shows sender info, path of delivery, timestamps, IP, etc.
  - Body → Contains the main message and any attachments.

## How Email Travels (Client → Server → Recipient)

- Many organizations use their own internal mail servers.
- Home users usually send emails through their ISP's mail server.
- Flow of an email:
  - a. Client sends email → goes to ISP/mail server
  - b. Server forwards the mail to receiver's mail server
  - c. Mail stays on receiver's server until the user downloads or views it
- Email servers run on Windows, Linux, Unix, etc., and manage message storage & transmission.

## **Email Investigation Perspective**

- Internal corporate emails are easier to trace because:
  - They follow UNC naming conventions
  - They use central authentication
  - Logs are well maintained → easy to find actual sender/receiver

## **Role of the Email Client**

- Displays the list of messages with:
  - Header
  - Sender name
  - Date & time
  - Size of message
- Allows the user to:
  - View
  - Compose
  - Reply
  - Delete messages



## Role of the Email Server

- Stores all user accounts and mailboxes.
- Each account has a text file/mail file to store messages.
- When a user clicks Send:
  - Client forwards email to the mail server
  - Server formats the info and appends it to the receiver's mailbox file

## Email Protocols Used (Important for Forensics)



### POP (Post Office Protocol)

- Stores only incoming emails
- Investigation usually done on the local workstation
- Emails are downloaded and often removed from server



### IMAP (Internet Message Access Protocol)

- Stores incoming + outgoing messages
- Copies may remain on:
  - Server
  - Workstation
  - Or both



## **MAPI (Microsoft Mail API)**

- Works similar to IMAP
- Common in MS Outlook/Exchange environments

## **HTTP**

- Used for web-based email clients like Gmail, Yahoo, Outlook Web
- Emails accessed through browser

## **SMTP (Simple Mail Transfer Protocol)**

- Responsible for sending and transferring emails
- Uses TCP port 25
- Easy to spoof, so attackers can send fake emails using this protocol

# Investigating E-mail Crimes and Violations

## Steps in E-mail Crime Investigation

1. Examine the email
2. Copy the message
3. Print the email
4. View the email headers
5. Examine the email headers
6. Check attachments
7. Trace the email origin

### 1. Examine the E-mail

- When an email-related crime is reported, the first step is to collect digital evidence.
- Evidence is usually taken from the victim's computer (the received email).
- Important steps:
  - Create a forensic image of the victim's hard drive.
  - Obtain system or email passwords if needed to access encrypted files.



- Take a printed copy of the suspicious email including headers.
- Note the sender's IP address from the header.

## **2. Copy the Email Message**

- Copy the email to a USB drive or forensic storage device.
- Ensure the copy is write-protected to maintain integrity.

## **3. Print the Email**

- Print the full email using the email client's print option.
- Printed copies are useful as legal evidence in court.

## **4. View the Full Mail Header**

Steps to view full header:

- Open the email
- Right-click → View Full Header
- Header file opens containing routing & sender details

## 5. Examine the Email Header

- Header contains:
- Sender's IP address (e.g., X-Originating-IP: [200.85.213.54])
- Return path
- Sender and receiver email IDs
- Mail servers involved in delivery
- Used to check authenticity and identify spoofing.

## 6. Examine Attachments

- If attachments exist:
- Copy them separately
- Print them if needed
- Scan for malware or hidden data
- Attachments may contain evidence like phishing pages, payloads, PDFs, etc.

## **7. Trace the Email Origin**

- The sender's IP address helps identify:
- Service provider
- Country/region
- Registered owner of the domain
- Attackers may use fake info, so verification is important.

### **Useful Websites for Tracing**

#### **www.arin.net**

- Finds domain owner from IP address
- Shows contact details registered with the domain

#### **www.freeality.com**

- Advanced search options: name, phone, email
- Allows reverse email search
- Helps identify real owner behind an email ID

**THANK  
YOU!!!**